Learning the Groove: Fine-Tuning Beat Tracking on the Jazz Trio Dataset

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ABSTRACT

Benchmarking beat and downbeat estimation in jazz poses unique challenges due to the genre's rhythmic fluidity, including syncopation, swing, expressive timing, improvisation, and frequent ambiguity in perceived groove. The Jazz Trio Dataset (JTD) provides a strong foundation for this task, offering approximately 44.5 hours of jazz trio recordings with detailed annotations for onsets, beats, and downbeats [1]. In this paper, we evaluate beat and downbeat tracking on JTD using two systems: madmom, a widely used MIR library based on recurrent neural networks and dynamic Bayesian network post-processing [2], and Beat This!, a recent state-of-the-art beat tracking architecture designed to achieve accurate tracking without DBN post-processing [3]. We report standard beat-tracking metrics using mir_eval [4], including F-measure, Cemgil accuracy, CMLc, CMLt, AMLc, and AMLt. We also implement a fine-tuning pipeline that improves Beat This! to the instrumentation and rhythmic vocabulary of jazz trio performance. Our results represent an improvement and highlight that jazz remains a challenging domain for general-purpose beat trackers, particularly in passages with sparse drum activity, ambiguous walking bass, expressive rubato, and strong off-beat comping. Fine-tuning on JTD improves domain adaptation by exposing the model to jazz-specific timing patterns. However, remaining errors suggest that beat tracking in jazz requires models that account not only for periodicity but also for style-specific musical cues and metrical interpretation.

1. INTRODUCTION

Automatic beat and downbeat tracking are central tasks in Music Information Retrieval (MIR), supporting applications such as music transcription, structure analysis, and computational musicology. In many popular music contexts, beat tracking systems perform well because rhythmic cues are often stable, repetitive, and reinforced by clear drum patterns. Jazz, however, challenges these assumptions. The beat may be implied rather than explicitly played, swing may intentionally push or pull against a strict grid, and rhythmic emphasis may shift fluidly between piano, bass, and drums. As a result, systems trained primarily on pop, rock, electronic, or metronomically stable datasets can struggle when applied to jazz trio recordings.

A live jazz trio performance is challenging due to its sparse yet rhythmically complex texture. The drummer may not explicitly mark the beat, the bassist's walking line only loosely implies pulse, and the pianist often emphasizes off-beats. As a result, beat-tracking models can produce octave errors, phase shifts, or locally correct but globally inconsistent interpretations. Downbeat tracking is even harder, since the first beat is not always clearly emphasized, as the genre itself is characterized by emphasis on the 2 and 4 of the bar, and most of the music's length is improvisations.

The first goal of this project is to evaluate and compare the performance of *madmom* and Beat This! on the Jazz Trio Dataset. *madmom*[1] is an open-source Python library for MIR and audio signal processing that includes established beat and downbeat tracking pipelines based on recurrent neural networks and temporal decoding [2]. Beat This![2] represents a newer approach that seeks high beat-tracking accuracy without relying on Dynamic Bayesian Network (DBN) post-processing [3]. The second goal is to fine-tune Beat This! on JTD to test whether domain-specific training improves beat and downbeat estimation for jazz trio recordings.

This paper makes three contributions. First, it provides a focused evaluation of two beat-tracking systems on jazz trio recordings. Second, it documents a reproducible pipeline for preparing JTD annotations, running inference, computing MIR evaluation metrics, and generating diagnostic click-track audio. Third, it proposes and implements a fine-tuning setup for adapting Beat This! to jazz-specific rhythmic conditions. The

[1] *https://github.com/cpjku/madmom*
[2] *https://github.com/CPJKU/beat_this*

implementation, evaluation scripts, and documentation are available at RobertTylman/MIR-Jazz-Beat-Tracking.

2. MODELS & DATASET

2.1 Beat and Downbeat Tracking

Beat tracking is the task of estimating the temporal locations of perceived beats in an audio signal, from the previous real-time digital signal processing method [5] to machine learning and deep learning methods.

Downbeat tracking extends this task by identifying the first beat of each measure, requiring not only pulse estimation but also metrical understanding. In common-time music, a system may correctly estimate the beat pulse while still assigning the wrong phase to the bar structure. Downbeat tracking is generally considered more challenging than beat tracking, as it requires higher-level musical understanding beyond periodicity, including harmonic changes, metrical structure, and long-term temporal dependencies[6]. For jazz, this distinction matters because the underlying beat can remain steady while surface accents and improvisational phrasing obscure the downbeat.

Evaluation is commonly performed by comparing predicted beat timestamps to reference annotations within a tolerance window. F-measure captures the balance between precision and recall, while Cemgil accuracy rewards predictions according to their temporal proximity to reference beats. Continuity-based metrics such as CMLc and CMLt measure whether a system maintains the correct metrical level and phase over time. AMLc and AMLt allow alternative metrical levels, which are useful for diagnosing subdivision mistakes, such as tracking half-time or double-time instead of the annotated beat.

2.2 The Jazz Trio Dataset

The Jazz Trio Dataset contains audio and annotations for jazz piano trio recordings, including beat, downbeat, and onset information [1]. Its focus only on piano, bass, and drums makes it particularly useful for studying beat tracking in a genre where rhythmic roles are distributed across instruments. Unlike datasets dominated by heavily produced popular music, JTD captures performance practices such as swing, comping, walking bass, drum ride patterns, breaks, and improvisational timing variation.

For this study, JTD serves both as an evaluation benchmark and as a fine-tuning corpus. Its annotations allow us to evaluate existing systems quantitatively, while its stylistic consistency makes it a suitable domain-specific dataset for adapting a general beat tracker to jazz.

2.3 madmom

madmom is a Python audio and MIR library that provides beat and downbeat tracking models based on neural network activation functions and temporal decoding [2]. In our experiments, we use the RNNBeatProcessor for beat estimation and the RNNDownBeatProcessor for joint beat/downbeat estimation. These models generate frame-level beat or downbeat activation functions, which are then converted into timestamp predictions using post-processing methods such as DBNs.

The strength of *madmom* lies in its stability and interpretability. Its temporal models encourage globally consistent beat sequences, thereby reducing spurious local predictions. However, this same reliance on learned priors and temporal smoothing can be limiting when the input contains stylistic timing patterns that differ from the training distribution. In jazz, *madmom* may lock onto a plausible pulse but choose the wrong metrical level, especially in passages with sparse percussion or ambiguous rhythmic emphasis.

2.4 Beat This!

Beat This! is a recent beat-tracking system designed to achieve accurate beat estimation without DBN post-processing [3]. Instead of relying on a separate probabilistic decoding stage to impose temporal consistency, the model is trained to produce predictions that are already temporally coherent. This design is attractive for jazz because it may reduce dependence on hand-tuned post-processing assumptions about tempo stability or metrical regularity. We adapt the JTD data format to match the expected input structure for Beat This!, run inference on the jazz recordings, and evaluate the resulting beat and downbeat predictions with the same *mir_eval* [2] metrics used for *madmom*.

3. METHODOLOGY

3.1 Dataset Preparation

We organized the Jazz Trio Dataset into paired audio and annotation files. Each audio track is associated with reference beat and downbeat timestamps. The annotation files are converted into a consistent format containing event times and beat labels. For beat-only evaluation, all annotated beat times are used as the reference sequence.

For downbeat evaluation, we extract the subset of annotated beats corresponding to the first beat of each measure.

Before evaluation, all files are checked for alignment between audio duration and annotation duration. Tracks with missing audio, malformed annotation files, or incompatible sampling assumptions are excluded from the evaluation set.

3.2 madmom Inference Pipeline

For *madmom* beat tracking, each audio file is passed through the *RNNBeatProcessor* to generate beat activation functions. These activations are then decoded into beat timestamps using *madmom's* beat tracking processor. For downbeat tracking, we use the *RNNDownBeatProcessor*, which estimates both beat and downbeat activations. The resulting predictions are formatted as timestamp sequences compatible with *mir_eval*. The *madmom* pipeline produces two outputs for each track: predicted beat timestamps and predicted downbeat timestamps. These are saved in CSV files and evaluated against the corresponding JTD annotations.

3.3 Beat This! Inference Pipeline

For Beat This!, JTD audio files are converted or referenced in the structure expected by the repository. We run *Beat This!* inference to generate predicted beat and downbeat timestamps for each track. As with *madmom*, predictions are saved in a standardized format so that we can evaluate across systems.

3.4 Fine-Tuning Beat This! on JTD

To tailor Beat This! to the rhythmic and stylistic characteristics of jazz trio performance, we fine-tuned the original pretrained model on the Jazz Trio Dataset as a domain-specific training corpus. The fine-tuning split was created from the 1,294 valid JTD tracks, where each valid track folder contained a *beats.csv* annotation file. To make the split reproducible, the track IDs were first sorted for determinism, then shuffled using a fixed random seed of 1337. The dataset was divided into a 70/15/15 train, validation, and test split, producing 906 training tracks, 194 validation tracks, and 194 test tracks. Only the training and validation tracks were written into Beat This!'s *single.split* file for fine-tuning, while the test tracks were kept separate and reserved only for final evaluation. This ensured that the model was evaluated on tracks it had not seen during training or checkpoint selection.

The fine-tuning process used the original Beat This! model weights as initialization. Input audio was transformed into the log-mel spectrogram representation expected by the model [3], and the beat and downbeat annotations were converted into target activation sequences. The model was then optimized to predict jazz-specific beat and downbeat activations from the input audio. After fine-tuning, model predictions were saved in a standardized format so that performance could be evaluated consistently across systems using *mir_eval*.

The purpose of fine-tuning was not only to improve aggregate beat-tracking scores, but also to examine whether exposure to jazz trio recordings helps the model learn timing cues that are more appropriate for this style. Existing beat-tracking models often require adaptation to specific musical domains [7]. In particular, we hypothesize that fine-tuning on JTD helped Beat This! better interpret beat-level timing information distributed across bass, drums, and piano comping, rather than relying primarily on the more regular rhythmic patterns often found in broader training data. However, while beat-tracking performance improved after fine-tuning, downbeat-tracking performance decreased, suggesting that the model became better at identifying local beat positions but struggled more with the higher-level metrical structure needed for accurate downbeat prediction.

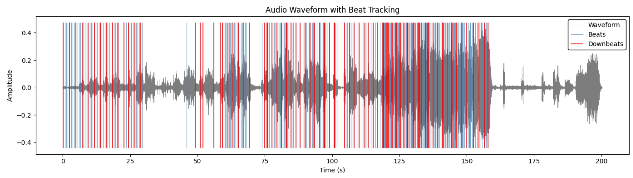

Figure 1. Analysis of beat and downbeat estimates on a jazz track from the test dataset.

3.5 Diagnostic Click-Track Rendering

In addition to quantitative evaluation, we generate diagnostic audio by overlaying metronome clicks onto the original recordings, as figure 1. This allows us to listen directly to model behaviour and identify musical conditions that produce errors. Listening to the predictions helps connect numerical errors to musical causes.

4. EVALUATION

4.1 Metrics

We evaluate beat and downbeat predictions using *mir_eval* [4]. The primary metric is F-measure, which compares predicted and reference events within a fixed temporal tolerance. We also report Cemgil accuracy, which assigns a continuous score based on the closeness of predicted events to reference events. To evaluate longer-term metrical consistency, we report CMLt and AMLt.

CMLt measures the total proportion of beats tracked correctly, allowing interruptions. AMLt are similar but allow alternative metrical levels, such as half-tempo or double-tempo interpretations. These metrics are especially important for jazz because a system may track a musically plausible pulse while disagreeing with the annotation's metrical level.

4.2 Initial Benchmark

The initial benchmarking results on the Jazz Trio Database (JTD) reveal a performance divergence between the transformer-based Beat This! architecture and the RNN-DBN framework of *madmom*. As shown in Table 1, *Beat This!* achieves a Beat F1 score of 0.9432, outperforming *madmom* (0.8702) by approximately 7.3%. This suggests that the attention mechanisms and convolutional-transformer hybrid in Beat This! are highly effective at navigating the micro-timing and swing nuances of jazz pulse detection without DBN.

However, the downbeat estimation metrics tell a different story. Both models show a decrease in F1 scores for downbeat identification, with *madmom* slightly outperforming *Beat This!* (0.6497 vs. 0.6189). This performance gap indicates the difficulty of downbeat tracking in jazz, where syncopated phrasing and frequent structural improvisations can obscure bar lines.

Model	Beat F1	Downbeat F1
Beat This! (BL)	0.9432	0.6189
madmom	0.8702	0.6497

Table 1. F1 scores across beat and downbeat estimations on *Beat This!* and *madmom*.

Model	Beat Cemgil	Downbeat Cengil
Beat This! (BL)	0.878	0.578
madmom	0.836	0.627

Table 2. Cemgil scores across beat and downbeat estimations on *Beat This!* and *madmom*.

Model	Beat CMLtl	Downbeat CMLt
Beat This! (BL)	0.832	0.530
madmom	0.609	0.510

Table 3. CMLt scores across beat and downbeat estimations on *Beat This!* and *madmom*.

Model	Beat AMLt	Downbeat AMLt
Beat This! (BL)	0.887	0.617
madmom	0.911	0.826

Table 4. AMLt scores across beat and downbeat estimations on *Beat This!* and *madmom*.

4.3 Experimental Conditions

We evaluate three conditions:

1. *madmom* baseline beat and downbeat tracking.
2. *Beat This!* baseline inference using the pretrained model.
3. *Beat This!* fine-tuned on JTD.

Each condition is evaluated on the same held-out test split. For each track, predictions are compared against JTD reference annotations. Track-level scores are then averaged to produce aggregate results. We also inspect per-track results to identify recordings where models perform especially well or poorly.

The results followed the expected trend for beat tracking but not for downbeat tracking (Figure 2). On the held-out JTD test split, Beat This! outperformed *madmom* on beat tracking, reflecting the advantage of its more modern architecture. The fine-tuned model reached a beat F-measure of 0.975, compared with 0.943 for the original Beat This! model and 0.870 for *madmom*. Beat Cemgil also improved from 0.887 to 0.929 after fine-tuning. These results suggest that fine-tuning helped the model learn jazz-specific beat-level timing cues, especially those distributed across bass, drums, and piano comping.

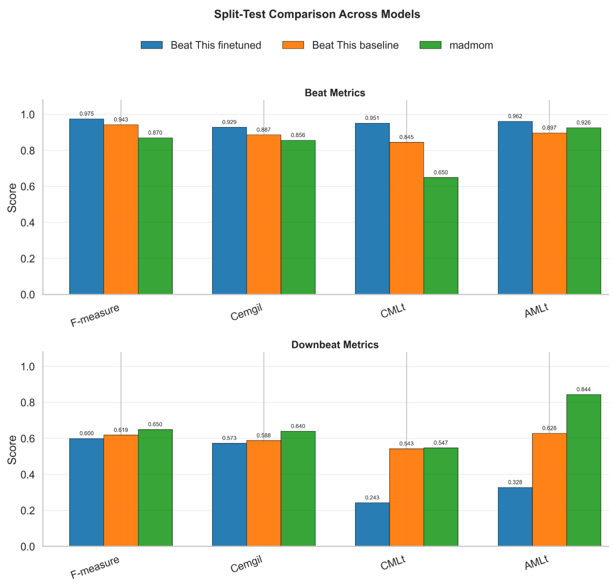

Figure 2. Evaluation metrics across beat and downbeat estimations on test split.

However, this improvement did not transfer to downbeat tracking. The fine-tuned model achieved a downbeat F-measure of 0.600, slightly below the original Beat This! model at 0.619 and *madmom* at 0.650. Its lower downbeat continuity scores suggest that fine-tuning improved local beat placement without improving the model's ability to infer higher-level metrical structure. In other words, the model became better at identifying where beats occur, but not necessarily better at determining which beats function

as bar-level downbeats. If AML scores are much higher than CML scores, this may indicate that the system is tracking a consistent but metrically shifted pulse.

4.4 Error Analysis

Our qualitative listening analysis suggests several recurring error categories. We focused on several of the lowest-performing tracks across the three models:

Track	Fine-Tuned BT	Baseline BT	Madmom
McCoy Tyner- *Darn That Dream*	0.531	0.468	0.504
Keith Jarrett- *I Thought About You*	0.619	0.375	0.412

Table 5. lowest-performing tracks across the three models.

In these examples, the tempo fluctuates noticeably, and the performances contain large dynamic changes that appear to affect beat-tracking stability. The models often struggle during sparse trio passages where the drums reduce explicit timekeeping or temporarily drop out. In these sections, the pulse is implied by bass motion, harmonic rhythm, or ensemble interaction rather than by clear percussive onsets. This is especially noticeable in *"I Thought About You*," where the drums are highly active but do not always establish a steady pulse. Similarly, changes in the double bass line, such as shifts into triplet figures or double-time motion, can cause the models to drift toward competing pulse interpretations.

This effect is particularly audible at the opening of *I Thought About You*, where the bassist plays a melodic, non-walking line rather than a conventional quarter-note pattern. This texture appears to disorient all three models early in the track. Swing timing also introduces ambiguity between beat-level and subdivision-level cues. Because swung eighth notes are unevenly spaced, models may interpret strong off-beat articulations as beat candidates. This can produce phase errors, especially when piano comping emphasizes syncopated positions.

It is also worth noting that both the baseline and fine-tuned Beat This! models often sound close to the beat, but systematically miss beat 2. The baseline model additionally produces more false positives, including audible double clicks around 00:20, while *madmom* remains more temporally consistent in that passage.

On *Darn That Dream*, *madmom* performs slightly better than the baseline Beat This! model, likely because the piece is more rhythmically stable. In contrast, the Beat This! predictions appear to alternate between half-time and double-time interpretations. This suggests that octave-level tempo errors remain a significant issue: the predicted tempo may be locally consistent, but it occurs at half or double the annotated metrical level.

Overall, downbeat tracking remains substantially harder than beat tracking. Jazz performances frequently avoid obvious downbeat reinforcement, especially during solos. Even when a model correctly follows the beat sequence, it may still assign the wrong bar position. This helps explain why fine-tuning improved beat-tracking performance while downbeat-tracking performance decreased. The fine-tuned model appears to have learned the local pulse of jazz trio performance more effectively than its larger-scale bar structure.

5. DISCUSSION

The comparison between *madmom* and *Beat This!* reflects a broader shift in MIR beat tracking. Earlier systems often combine neural activation models with explicit temporal decoding, while newer systems increasingly attempt to learn temporal structure directly. For jazz, both approaches have advantages. DBN-style post-processing can enforce continuity, which helps prevent erratic predictions. However, if the assumed tempo or meter model does not match the performance, the decoder can stabilize the wrong interpretation.

The fine-tuning results show that Beat This! can benefit from exposure to stylistically relevant jazz trio recordings, especially for beat-level prediction. The large improvement in beat F-measure suggests that the model learned more appropriate local timing cues from JTD. However, the decline in downbeat performance shows that improved beat tracking does not automatically imply improved metrical understanding. Downbeat estimation requires the model to infer bar position from longer-range musical context, which may be especially difficult in jazz trio performances where downbeats are often implied rather than explicitly emphasized.

Overall, fine-tuning made Beat This! substantially better at the core beat-tracking task, but it did not improve, and may have weakened downbeat estimation. This suggests that future work should treat beat and downbeat tracking as related but distinct problems, potentially requiring different training objectives, stronger metrical supervision, or decoding strategies that preserve bar-level structure while still allowing flexibility for jazz timing.

6. LIMITATIONS

This study has several limitations. First, the conclusions depend on the specific train, validation, and test split used in the experiments. Although the held-out test split was kept separate from fine-tuning, any reported improvement should be interpreted carefully and ideally validated across additional splits. In particular, future evaluations should further control for possible recording session, artist, or album leakage to ensure that performance gains reflect generalization rather than memorization of closely related material.

Second, the current evaluation relies on standard MIR beat-tracking metrics. These metrics are useful for comparing systems, but they may not fully capture perceptual correctness in jazz performance. For example, a prediction that follows a musically plausible half-time or double-time feel may be penalized heavily even if a listener would understand it as rhythmically coherent.

Third, the analysis is limited to jazz trio recordings. Results may not generalize to larger jazz ensembles, solo piano, vocal jazz, fusion, or other rhythmically flexible styles. The cues available in bass, drums, and piano trio performance may differ substantially from those in other jazz settings.

Fourth, the fine-tuning process depends on the annotation quality and consistency of JTD. Beat and downbeat annotations encode one interpretation of the performance, but ambiguous passages may allow multiple valid metrical readings. This is especially important for jazz, where swing feel, syncopation, implied time, and expressive timing can make the “correct” beat or downbeat less perceptually obvious than in more regular musical styles.

7. CONCLUSION

Jazz tracking remains a challenging problem for MIR because the beat is often implied through interaction, timing feel, and stylistic convention rather than stated through repetitive percussive cues. In this paper, we evaluated *madmom* and Beat This! on the Jazz Trio Dataset and developed a pipeline for fine-tuning Beat This! on jazz-specific data. The evaluation framework uses standard mir_eval metrics alongside diagnostic click-track rendering to connect quantitative scores with musical error types.

Our analysis suggests that modern architectures such as Beat This! are promising for jazz beat tracking, and that fine-tuning on JTD can improve adaptation to the genre’s rhythmic vocabulary. At the same time, persistent errors in sparse textures, swung subdivisions, downbeat identification, and octave-level pulse selection show that jazz remains a demanding benchmark. Continued progress will likely require models that understand not only periodic timing but also the musical roles of instruments, ensemble interaction, and metrical ambiguity.

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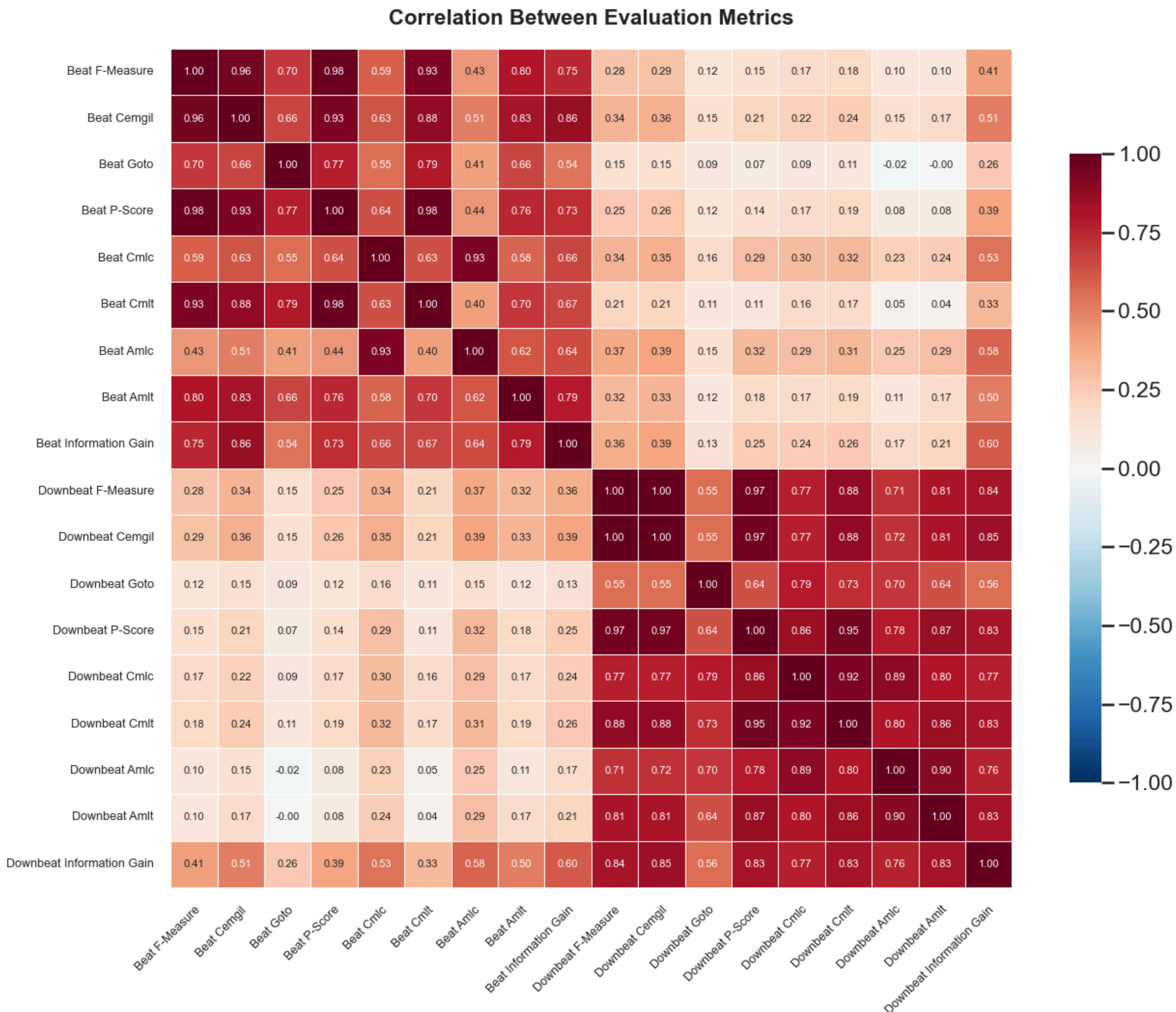

Appendix Chart: Full metrics analysis of Finetuned Beat This!